

Paper 9: Logic Waves and Half-Wavelength Censorship

The Odd-Harmonic Ladder, Amplitude-Free Coherence Conditions, Kinematic Stability of Composites, and the Existence Ceiling for Single Entities

Author: Noriaki Kihara

Affiliation: WF System Co., Ltd.

ORCID: 0009-0004-6753-4020

Version: v0.2

Date: June 2026

DOI (this version): 10.5281/zenodo.20640463

Concept DOI: 10.5281/zenodo.20640462

License: CC BY 4.0

Abstract

In this paper, we verify the **existence conditions and stability** of composite states (clusters consisting of many cells) in the reciprocal dual model by explicit construction of interference patterns.

First, **consolidation of the logic wave hypothesis**: the per-axis value of the counting condition, $|k_i| + 1/2$, is an odd-harmonic sequence with the zero point $1/2$ as its base, and a spectrum containing only odd harmonics is the fingerprint of a binary symmetric signal (square wave, logic wave). That the phase wave carries no amplitude is not an approximation but a **coherence condition** demanded both by the theorem layer (the integer partition structure) and by curvature consistency. Information resides in edge positions, i.e., in the relative phases of the harmonic ladder, and a parallel translation of a square pulse is exactly equivalent to a linear phase multiplication on the coefficients (position = phase).

Second, **kinematic stabilization by half-wavelength censorship**: the maximum of the curvature-induced anharmonic shift of the ladder is exactly $1/2$ (the lowest mode, verified in rational arithmetic), and the censorship condition “shift $\leq$ half the resolution” is satisfied **exactly at the limit**. Without censorship, if the shift is taken as real dynamics, a composite dephases and collapses within 3–6 fundamental periods; under censorship, the continuous decay channel is kinematically closed, and the only remaining decays are the discrete channels (Paper 7). The stability of composite particles is not a dynamical accident but a **quantization of the deformation space**. As a by-product, we observe that the maximum dimension in which this protection holds is $d = 4$ (the fixing of normalization is deferred to Paper 11).

Third, **interference closure tests and the existence ceiling**: a single cell can be constructed exactly from odd harmonics alone (width/period = $1/2$ is the unique filling fraction at which the even harmonics vanish identically). The condition for odd-harmonic ladders to nest is that the scale ratio be odd (the **odd nesting theorem** — a wave-level rederivation of the odd- k rule of the genealogy). Computing, by exact Fourier analysis over the entire odd- s ladder, whether an occupancy structure can be certified by its own waves alone (the eye opening in the fundamental-wave band), we have established that the stable species $s = 1, 3, 5$ certify themselves with strong margins ($\geq +0.22$), the decay threshold $s = 7$ cannot be certified, $s \geq 9$ is critical, and **for $s \geq 25$ certification at the composite scale disappears, while for $s \geq 49$ certification fails in every band of the odd sector** (verified down to band depth 31; the margin deteriorates monotonically with s). A control experiment shows that allowing even harmonics restores certification, so the **closure is intrinsic to the logic wave (odd, binary) structure**.

As a consequence, a large content S cannot exist as a flat single entity, but only as a nesting of certifiable small entities — **hierarchization is not a choice but a compulsion**.

Keywords

square wave, odd harmonics, Gibbs phenomenon, Nyquist orthogonality, half-wavelength censorship, kinematic stability, soliton-like wave packet, existence ceiling, hierarchization, self-certification

1. Introduction

The stable species $\{1, 3, 5\}$ and the decay threshold $s = 7$ established in Paper 7 [4] were obtained from the **counting** of channels. The question of this paper is whether that stability is supported at the level of waves:

With sharp-edged odd-harmonic waves, (1) can the interference pattern of a stable 4-dimensional cell actually be drawn? (2) When 9 of them, or 137 of them, are stacked up, can a stable interference pattern be drawn? Without examining this, neither the **limit** up to which stability can be maintained nor the **reason** for stability can be explained.

The toolkit is owed to the classics of signal theory: the odd-harmonic expansion of the square wave and the Gibbs phenomenon accompanying truncation [5], and the sampling of band-limited signals and transmission free of intersymbol interference (Nyquist/Shannon [6]). The architecture of the stability argument shares its logic with the stabilization of the atom, in which continuous decay

channels are closed by quantization (Bohr [7]), and the aspect of a localized wave that preserves its shape corresponds structurally to solitons (Zabusky & Kruskal [8]). All of these are structural correspondences, not identifications.

2. The Logic Wave Hypothesis

2.1 The odd harmonics live inside the lattice

The per-axis value of the counting condition is $|k_i| + \frac{1}{2} = (2|k_i| + 1)/2 \in \{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots\}$, that is, an **odd-harmonic sequence with the zero point $1/2$ as its base** (even multiples do not appear). Since a signal with half-wave symmetry $f(t + T/2) = -f(t)$ has exactly no even harmonics, an odd-harmonic spectrum is the **fingerprint of a binary symmetric signal (a logic wave)**.

2.2 The harmonic ladder = the hierarchical genealogy

The parent-child relation of the cascade is, in representative frequencies, parent $\nu = k \times$ child ν' (with k odd). Therefore, the odd-harmonic sequence of a given fragment is precisely the **ancestral chain** of that fragment. The closure of mixed- k cascades realizes exactly the complete odd-harmonic sequence (odd numbers = products of odd primes).

2.3 The absence of amplitude is a coherence condition

That the phase wave carries no amplitude follows from two independent requirements.

1. **Integrity of the theorem layer:** the theorems of Papers 5–8 (the splitting theorem, the freezing theorem, Z_2 , tiling) all stand on the integer combinatorial structure of “odd partitions of S .” If continuous amplitude were a degree of freedom, the bookkeeping would dissolve into the continuum and not a single theorem would survive.
2. **Curvature consistency:** amplitude is a displacement outward from the direction of propagation, and the local energy density of its square couples to the local curvature and self-interacts. A phase-only wave (taking values ± 1) has uniform local energy density, so the linear phase computation survives exactly. Note that a 0/1 logic wave, by idempotency $\chi^2 = \chi$, has its squared density localized on itself, so the influence of curvature is unavoidable; its consequence (anharmonicity) is the subject of §3.

2.4 Position = phase

All the information of a square wave lies in its edge positions (zero crossings); the height carries no information. By the Fourier shift theorem, a parallel translation of a square pulse is exactly equivalent to a linear phase multiplication on the coefficients. **The information called “position” lives not in amplitude**

but in the relative phases of the harmonic ladder. Truncating the ladder at $\nu_{\max}$ gives an edge width $\sim 1/(2\nu_{\max})$, and the ringing, about 9%, is immortal regardless of the truncation order (Gibbs [5]). The zero point $\delta_{\min}$ reappears as the minimal edge width = resolution limit.

A cell is exactly an odd-harmonic (half-wave) object; truncation gives Gibbs edges

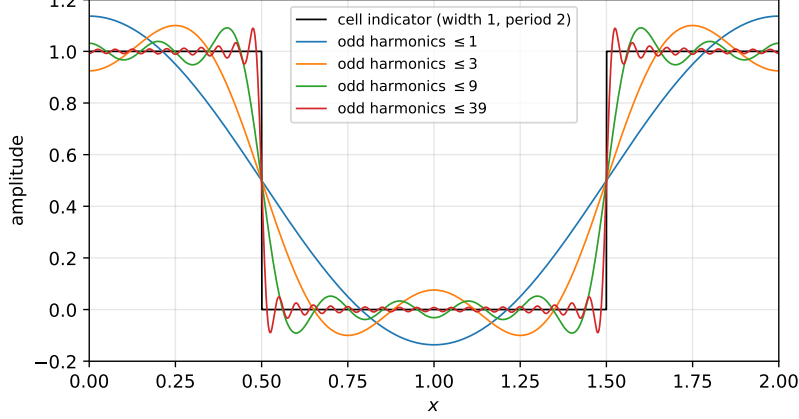


Figure 1: Figure 1. The cell as an exact odd-harmonic object (Gibbs edges)

Figure 1. The cell as a native inhabitant of the odd harmonics: the indicator function of width 1 on period 2 (even harmonics are exactly zero) and the Gibbs edges of odd truncated partial sums.

3. Half-Wavelength Censorship: Kinematic Stabilization of Composites

3.1 Exact bound on the anharmonic shift

The ladder on the container of a composite becomes anharmonic due to curvature. From the identity $(\ell + \frac{3}{2})^2 - \ell(\ell + 3) = \frac{9}{4}$, the maximal shift is

$$\Delta\nu_\ell = \frac{9/4}{(\ell + \frac{3}{2}) + \sqrt{\ell(\ell + 3)}}, \quad \Delta\nu_1 = \frac{9/4}{9/2} = \boxed{\frac{1}{2}}$$

which is exactly monotonically decreasing in ℓ , and **its maximum coincides exactly with the zero point 1/2 at the lowest mode** (verified in rational arithmetic). The censorship condition “distortion shift $\leq$ half the resolution” is satisfied not approximately but **exactly at the limit**.

3.2 Collapse time without censorship (reductio ad absurdum)

If the shift is taken as a real dynamical phase drift, the fidelity of a ladder with square-wave-type weights **falls below 50% within 3–6 fundamental periods** (verified for ladders of 2–8 rungs). Just as a classical electron falls into the nucleus in 10^{-11} seconds by continuous radiation, a composite would collapse in an instant if the continuous channel were open.

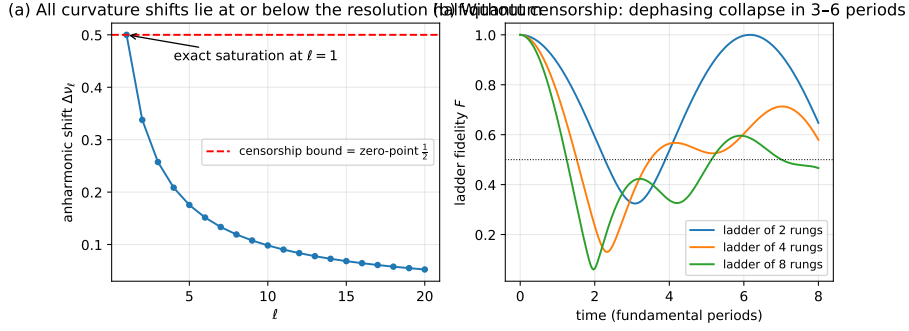


Figure 2: Figure 2. Censorship: exact bound on anharmonic shifts and dephasing without it

Figure 2. (a) Exact bound on anharmonic shifts: every shift is at most the zero point $1/2$, with equality only at $\ell = 1$. (b) Dephasing collapse without censorship (3–6 fundamental periods).

3.3 Stability under censorship

All shifts are at most half the resolution (the unit record, Paper 6 [3]), and a difference that is not recorded has no fact (the record theorem). Therefore the continuous decay channel is **kinematically closed**, and the only remaining decays are the discrete channels — coherent splitting (odd partitions + Z_2 + the capacity law, Paper 7).

	Atom (real physics)	Composite in this model
Continuous decay channel	Continuous radiation of the classical electron	Curvature dephasing (3–6 periods)
Quantum closing the channel	$\hbar/2$	$\delta_{\min}^2 = 1/2$
Remaining transitions	Quantum transitions between discrete spectra	Discrete channels (enumerated in Paper 7)

The stability of composites is not a dynamical accident but a **quantization of the deformation space** (a structural correspondence with the Bohr model [7]; no identification is made).

3.4 The cloud of the composite and the three-layer description

From the band structure, nothing inside a composite oscillates faster than the composite itself (the composite is its own ultraviolet cutoff). The internal configuration is B_4 gauge (Paper 6), and the observable content is exhausted by the gauge-invariant multiplets and the representative label ($R = 3$: cell basis $137 \rightarrow$ multiplets $7 \rightarrow$ representative 1). The decay of an individual cell is gauge non-invariant and is not defined as a physical event — the attack surface is far smaller than the nominal capacity.

3.5 An observation concerning dimension

The maximal shift of the S^d ladder reduces exactly to $\Delta\nu_1(d) = (\sqrt{d} - 1)^2/2$, the censorship condition $\leq 1/2$ is equivalent to $d \in [0, 4]$, and saturation occurs only at the two endpoints. $d = 4$ is the maximum dimension in which this protection holds. The status of this observation (the fixing of normalization and coupling) is treated in Paper 11.

4. Interference Closure Tests: The Ceiling of Existence

4.1 Exact closure of the single cell

The indicator function of a cell of width 1 placed on period 2 makes the even harmonics zero (to machine precision) — width/period = $1/2$ is the **unique filling fraction** at which half-wave symmetry holds exactly, and the cell is a native inhabitant of the odd harmonics. Already in the minimal construction (one fundamental wave per axis), the edge width 0.43 falls inside the zero-point width $1/2$.

4.2 The odd nesting theorem

The condition for the ladder of a cell (frequencies $m/2$, m odd) to ride on the ladder of a composite ($n/2R$, n odd) is $n = mR$, that is, **the scale ratio R must be odd** (the retention rate is 100% for odd R and exactly 0% for even R). The odd- k rule of the genealogy (§2.2) is rederived from wave closure alone. A composite with even R cannot carry a single coherent logic wavefront — by a route independent of the R^2 classification theorem (Paper 5 [2]: single states are odd), we arrive at the same exclusion.

4.3 Fundamental-wave certifiability and the three-band structure

We judged whether a composite “can certify its own occupancy structure using only waves of its own wavelength,” using the exact Fourier coefficients of the occupancy field (structure factor $\times$ sinc, no lattice jitter) and the eye opening in the odd bands (separation of occupied/empty slots). The best opening

under the composite-scale criterion (wavelength of the certifying band $\geq$ the cell fundamental wavelength):

s	1	3	5	7	9	11	13	17	21	25	33	49	81	121
best	+1.64	+0.29	+0.23	-0.10	+0.29	+0.33	+0.06	+0.12	+0.03	-0.04	-0.09	-0.07	-0.20	-0.37
eye														

Three-band structure: - **Certified band** $s \in \{1, 3, 5\}$: margin $\geq +0.23$ with a single fundamental wave. Coincides with the absolutely stable species of Paper 7. - **Critical band** $s = 9-21$: certifiable, but the margin decays and oscillates. The exception is $s = 7$ — the unique small label that cannot be certified at the composite scale (coinciding with the decay threshold). - **Forbidden band** $s \geq 25$: composite-scale certification disappears. For $s \geq 49$, certification fails in **every band** of the odd sector (verified down to band depth 31, wavelength 0.45 cells, saturating at -0.07), and the margin deteriorates monotonically with s (-0.37 at $s = 121$).

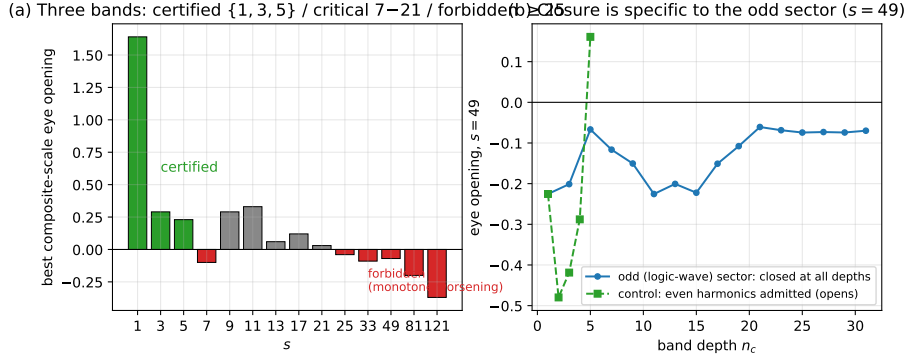


Figure 3: Figure 3. Three-band certification structure and odd-sector closure

Figure 3. (a) The three-band structure of fundamental-wave certifiability (exact values): certified band $\{1, 3, 5\}$ / critical band / forbidden band. (b) The odd sector of $s = 49$ is closed down to depth 31, while the control experiment (even harmonics allowed) is open — the closure is intrinsic to logic waves.

4.4 Control experiment: the closure is intrinsic to logic waves

In the full band with even harmonics also allowed, $s = 25$ opens at depth 4 and $s = 49$ at depth 5. The occupancy information itself exists, but it **can live only in the even harmonics** — and the even harmonics lie outside the alphabet of binary symmetric signals. Large clusters are not “undrawable”; they are “undrawable as logic waves.”

4.5 The compulsion of hierarchization

$s = 81$ cannot exist as a single entity, but as a nesting of nine children of $s = 9$ it can be composed entirely of entities certifiable at every level.

A large S cannot exist as a flat cluster, but only as a nesting of certifiable small entities. **The inward self-similar hierarchy (Paper 4 [1]) is neither a choice nor a tendency, but a compulsion under the logic wave ontology.** That “clusters of high frequency cannot exist stably” is shown, without requiring any dynamics of decay, as a kinematic impossibility of the mode of existence.

Seen from outside, a stable species takes the form of a quasi-square solitary wave packet that carries a built-in odd comb of spacing $1/R'$, is localized within an extent $2R'$, and restores itself every period (a minimal-uncertainty wave packet with half-width $\times$ fundamental wave = $1/2$). Whereas the classical soliton [8] cancels dispersion by nonlinearity, the wave packet of this model keeps its shape because **copyright kills dispersion** — a structural correspondence).

5. Scope of Claims of This Paper

What we claim: (1) Lattice = odd-harmonic ladder, ladder = genealogy. (2) Absence of amplitude = coherence condition (two independent requirements). (3) The exact bound $1/2$ on the anharmonic shift and kinematic stabilization by copyright (including the reductio). (4) Exact closure of the single cell and the odd nesting theorem. (5) The three-band structure of fundamental-wave certifiability and the existence ceiling ($s \geq 25/49$), the intrinsicality to logic waves shown by the control experiment, and the compulsion of hierarchization.

What we do not claim: (1) Identification with matter, fields, or particles in the real universe (“logic wave,” “stability,” and “wave packet” are descriptions of modes of construction within the model). (2) Dynamics of decay (transition rates, lifetimes). (3) Promotion of the $d = 4$ statement (it remains an observation; the fixing of the conditions is for Paper 11). (4) A complete description of the longitudinal direction of the harmonic ladder (inter-level coherence) — the energy bookkeeping per harmonic (truncation solution or genealogy solution) is explicitly left as an open problem.

6. Conclusion

The stability of composite states is supported by a two-tier structure: the counting tier (Paper 7: the presence or absence of channels) and the wave tier (this paper: copyright closes continuous decay, and interference closure draws the ceiling of existence). The two are independent computations, yet they agree

on the stable species $\{1, 3, 5\}$ and the threshold $s = 7$. Moreover, the wave tier supplied information absent from the counting — the existence ceiling and the compulsion of hierarchization. The mode of existence as a single gigantic entity is, in this model, forbidden by the logic wave structure itself.

Appendix: Essentials of the Reproduction Procedure

The anharmonic shifts rest on the identity and rational arithmetic; the collapse times on numerical integration of ladder fidelity; the closure tests on exact Fourier coefficients (structure factor $\times$ sinc) and exhaustive evaluation of eye openings ($s \leq 121$, including robustness checks at multiple resolutions). The scripts and research notes are provided in the public repository (<https://github.com/WurabeSeiji/ai-chat-logs-open>).

References

- [1] Noriaki Kihara, Paper 4, v1.0, 2026. Concept DOI: 10.5281/zenodo.20638962.
 - [2] Noriaki Kihara, Paper 5, v0.2, 2026. Concept DOI: 10.5281/zenodo.20640454.
 - [3] Noriaki Kihara, Paper 6, v0.2, 2026. Concept DOI: 10.5281/zenodo.20640456.
 - [4] Noriaki Kihara, Paper 7, v0.2, 2026. Concept DOI: 10.5281/zenodo.20640458.
 - [5] E. Hewitt and R. E. Hewitt, “The Gibbs–Wilbraham phenomenon: An episode in Fourier analysis,” *Archive for History of Exact Sciences* 21 (1979), 129–160.
 - [6] C. E. Shannon, “Communication in the presence of noise,” *Proceedings of the IRE* 37 (1949), 10–21.
 - [7] N. Bohr, “On the constitution of atoms and molecules,” *Philosophical Magazine* 26 (1913), 1–25.
 - [8] N. J. Zabusky and M. D. Kruskal, “Interaction of ‘solitons’ in a collisionless plasma and the recurrence of initial states,” *Physical Review Letters* 15 (1965), 240–243.
-

License

This paper is published under CC BY 4.0.
Reuse, adaptation, translation, and citation are permitted, provided that the author name, version, publication date, and source are clearly indicated.